

# Netflix Content Strategy Analysis: Trends, Patterns & Recommendations

## Data Analysis Project

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## Project Overview

This project presents an end-to-end data analysis of Netflix's global content catalog, covering both Movies and TV Shows. Using Python, the dataset was cleaned, explored, and visualized to uncover patterns across content type, country of origin, genre, audience rating, and release timing. Each analysis is paired with clear observations and business insights, culminating in data-driven recommendations aimed at supporting Netflix's content strategy decisions. The project demonstrates a complete analytics workflow — from raw data to business-ready conclusions — using Pandas, NumPy, Matplotlib, and Seaborn.

## Business Objective

This project evaluates Netflix's content catalog to uncover how the platform's library has evolved across content type, geography, genre, audience rating, and release timing. By examining these patterns through structured data analysis, the study aims to surface actionable insights into Netflix's content strategy and offer evidence-based recommendations to guide future acquisition, production, and release decisions.

## Tools Used

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn

## Dataset

Netflix Movies and TV Shows Dataset

# Stakeholders

- Content Team
- Business Managers
- Marketing Team
- Strategy Team

# KPIs

- Total Content
- Total Movies
- Total TV Shows
- Top Countries
- Top Genres
- Average Movie Duration
- Most Common Rating
- Content Growth Over Time

# Business Questions

1. What is the overall composition of Netflix's catalog — how many Movies versus TV Shows are there, and what proportion does each represent?
2. Which countries contribute the most content to Netflix, and how does this ranking change when titles produced through international collaborations are accounted for individually?
3. How has Netflix's content library grown over time, and are there specific years or months when content additions peak?
4. What are the most common content ratings on Netflix, and what does this suggest about the platform's primary target audience?
5. Which genres are most represented in Netflix's catalog, and what does this reveal about the platform's content focus?
6. What is the typical duration of Netflix Movies, and how many seasons do most TV Shows have?

# Importing Libraries and Dataset

with Data Cleaning Process

```
# Import Libraries
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt # type: ignore
import seaborn as sns # type: ignore
```

```
# Display plots inside Jupyter Notebook
%matplotlib inline
```

```
# Load Dataset
```

```
df = pd.read_csv("C:/Users/anura/Downloads/netflix_titles.csv")
```

```
df.head()
```

|   | show_id | type    | title                 | director        | cast                                              | country       | date_added         |
|---|---------|---------|-----------------------|-----------------|---------------------------------------------------|---------------|--------------------|
| 0 | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson | NaN                                               | United States | September 25, 2021 |
| 1 | s2      | TV Show | Blood & Water         | NaN             | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... | South Africa  | September 24, 2021 |
| 2 | s3      | TV Show | Ganglands             | Julien Leclercq | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... | NaN           | September 24, 2021 |
| 3 | s4      | TV Show | Jailbirds New Orleans | NaN             | NaN                                               | NaN           | September 24, 2021 |
| 4 | s5      | TV Show | Kota Factory          | NaN             | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... | India         | September 24, 2021 |

```
df.tail()
```

|      | show_id | type  | title  | director      | cast                                              | country       | date_ad            |
|------|---------|-------|--------|---------------|---------------------------------------------------|---------------|--------------------|
| 8802 | s8803   | Movie | Zodiac | David Fincher | Mark Ruffalo, Jake Gyllenhaal, Robert Downey J... | United States | Novemb<br>20, 2019 |

|             | show_id | type    | title       | director        | cast                                              | country       | date_ad           |
|-------------|---------|---------|-------------|-----------------|---------------------------------------------------|---------------|-------------------|
| <b>8803</b> | s8804   | TV Show | Zombie Dumb | NaN             | NaN                                               | NaN           | July 1, 2         |
| <b>8804</b> | s8805   | Movie   | Zombieland  | Ruben Fleischer | Jesse Eisenberg, Woody Harrelson, Emma Stone, ... | United States | Novemb<br>1, 2019 |
| <b>8805</b> | s8806   | Movie   | Zoom        | Peter Hewitt    | Tim Allen, Courteney Cox, Chevy Chase, Kate Ma... | United States | January 2020      |
| <b>8806</b> | s8807   | Movie   | Zubaan      | Mozez Singh     | Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan... | India         | March 2 2019      |

```
df.shape
```

```
(8807, 12)
```

```
df.columns
```

```
Index(['show_id', 'type', 'title', 'director', 'cast', 'country',  
'date_added',  
      'release_year', 'rating', 'duration', 'listed_in', 'description'],  
      dtype='object')
```

```
df.dtypes
```

```
show_id      object  
type         object  
title        object  
director     object  
cast         object  
country      object  
date_added   object  
release_year  int64  
rating       object  
duration     object  
listed_in    object  
description   object  
dtype: object
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        6173 non-null   object
4   cast            7982 non-null   object
5   country         7976 non-null   object
6   date_added      8797 non-null   object
7   release_year    8807 non-null   int64
8   rating          8803 non-null   object
9   duration        8804 non-null   object
10  listed_in       8807 non-null   object
11  description     8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
df.describe()
```

|       | release_year |
|-------|--------------|
| count | 8807.000000  |
| mean  | 2014.180198  |
| std   | 8.819312     |
| min   | 1925.000000  |
| 25%   | 2013.000000  |
| 50%   | 2017.000000  |
| 75%   | 2019.000000  |
| max   | 2021.000000  |

```
df.describe(include="object")
```

|        | show_id | type  | title                | director      | cast               | country       | date_ad      |
|--------|---------|-------|----------------------|---------------|--------------------|---------------|--------------|
| count  | 8807    | 8807  | 8807                 | 6173          | 7982               | 7976          | 8797         |
| unique | 8807    | 2     | 8807                 | 4528          | 7692               | 748           | 1767         |
| top    | s1      | Movie | Dick Johnson Is Dead | Rajiv Chilaka | David Attenborough | United States | January 2020 |
| freq   | 1       | 6131  | 1                    | 19            | 19                 | 2818          | 109          |

```
df.isnull().sum()
```

```
show_id      0
type         0
title        0
director    2634
cast        825
country     831
date_added   10
release_year  0
rating       4
duration     3
listed_in    0
description  0
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df["type"].unique()
```

```
array(['Movie', 'TV Show'], dtype=object)
```

```
df["rating"].unique()
```

```
array(['PG-13', 'TV-MA', 'PG', 'TV-14', 'TV-PG', 'TV-Y', 'TV-Y7', 'R',
      'TV-G', 'G', 'NC-17', '74 min', '84 min', '66 min', 'NR', nan,
      'TV-Y7-FV', 'UR'], dtype=object)
```

```
df["country"].nunique()
```

```
748
```

```
missing_percentage = (df.isnull().sum() / len(df)) * 100
```

```
missing_percentage.sort_values(ascending=False)
```

```
director      29.908028
country       9.435676
cast          9.367549
date_added    0.113546
rating        0.045418
duration      0.034064
show_id       0.000000
type          0.000000
title         0.000000
release_year  0.000000
listed_in     0.000000
description    0.000000
dtype: float64
```

```
# Create a copy of the original dataset
```

```
df_clean = df.copy()
```

```
df_clean.duplicated().sum()
```

```
np.int64(0)
```

```
# Fill Missing Director Values
```

```
df_clean["director"] = df_clean["director"].fillna("Unknown")  
df_clean["director"].isnull().sum()
```

```
np.int64(0)
```

```
# Fill Missing Cast
```

```
df_clean["cast"] = df_clean["cast"].fillna("Unknown")  
df_clean["cast"].isnull().sum()
```

```
np.int64(0)
```

```
# Fill Missing Country
```

```
df_clean["country"] = df_clean["country"].fillna("Unknown")  
df_clean["country"].isnull().sum()
```

```
np.int64(0)
```

```
# Drop Rows with Missing Date, Rating, or Duration
```

```
df_clean = df_clean.dropna(subset=["date_added", "rating", "duration"])  
df_clean.isnull().sum()
```

```
show_id      0  
type         0  
title        0  
director     0  
cast         0  
country      0  
date_added   0  
release_year 0  
rating       0  
duration     0  
listed_in    0  
description   0  
dtype: int64
```

```
# Check Dataset Shape Again
```

```
df_clean.shape
```

```
(8790, 12)
```

```
# Convert date_added to Datetime
```

```
df_clean["date_added"].dtype
```

```
df_clean["date_added"] = pd.to_datetime(df_clean["date_added"],
                                         format="mixed")
df_clean["date_added"].dtype

dtype('<M8[us]')

# Save the Cleaned Dataset

df_clean.to_csv("../data/netflix_cleaned.csv", index=False)
```

## Data Cleaning Summary

- Created a copy of the original dataset.
- No duplicate records found.
- Filled missing values in director, cast, and country with "Unknown".
- Removed rows with missing values in date\_added, rating, and duration.
- Converted date\_added to datetime format.
- Saved the cleaned dataset for further analysis.

```
df_clean.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        6173 non-null   object
4   cast            7982 non-null   object
5   country         7976 non-null   object
6   date_added      8797 non-null   object
7   release_year    8807 non-null   int64
8   rating          8803 non-null   object
9   duration        8804 non-null   object
10  listed_in       8807 non-null   object
11  description      8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
# Final verification of cleaned dataset
```

```
print("=" * 50)
print("Shape:", df_clean.shape)
print("=" * 50)

print("\nMissing Values:")
print(df_clean.isnull().sum().sum())

print("\nDuplicate Rows:")
print(df_clean.duplicated().sum())
```



```

print("\nData Types:")
print(df_clean.dtypes)

=====
Shape: (8790, 12)
=====

Missing Values:
0

Duplicate Rows:
0

Data Types:
show_id                str
type                  str
title                 str
director              str
cast                  str
country               str
date_added             datetime64[us]
release_year           int64
rating                 str
duration               str
listed_in              str
description             str
dtype: object

```

## Exploratory Data Analysis

With the dataset cleaned and prepared, this section explores Netflix's content catalog in depth. Each analysis addresses one of the business questions outlined earlier, examining patterns across content type, country, release timing, genre, rating, and duration. Visualizations are used throughout to make trends easier to interpret, followed by observations and business insights derived from each finding.

*# How many total titles are available on Netflix?*

```

total_content = df_clean.shape[0]
print(f"Total Content on Netflix: {total_content}")

```

Total Content on Netflix: 8790

*# How many Movies and TV Shows are available?*

```

content_count = df_clean["type"].value_counts()
content_count

```

```

type
Movie      6126
TV Show    2664
Name: count, dtype: int64

```

```
# What percentage of the catalog is Movies vs TV Shows?
```

```
count_percentage =  
    (df_clean["type"].value_counts(normalize=True).mul(100).round(2))  
count_percentage
```

```
type  
Movie      69.69  
TV Show    30.31  
Name: proportion, dtype: float64
```

```
# First Visualization - Count Plot
```

```
plt.figure(figsize=(8,5))
```

```
ax = sns.countplot(data=df_clean, x="type")
```

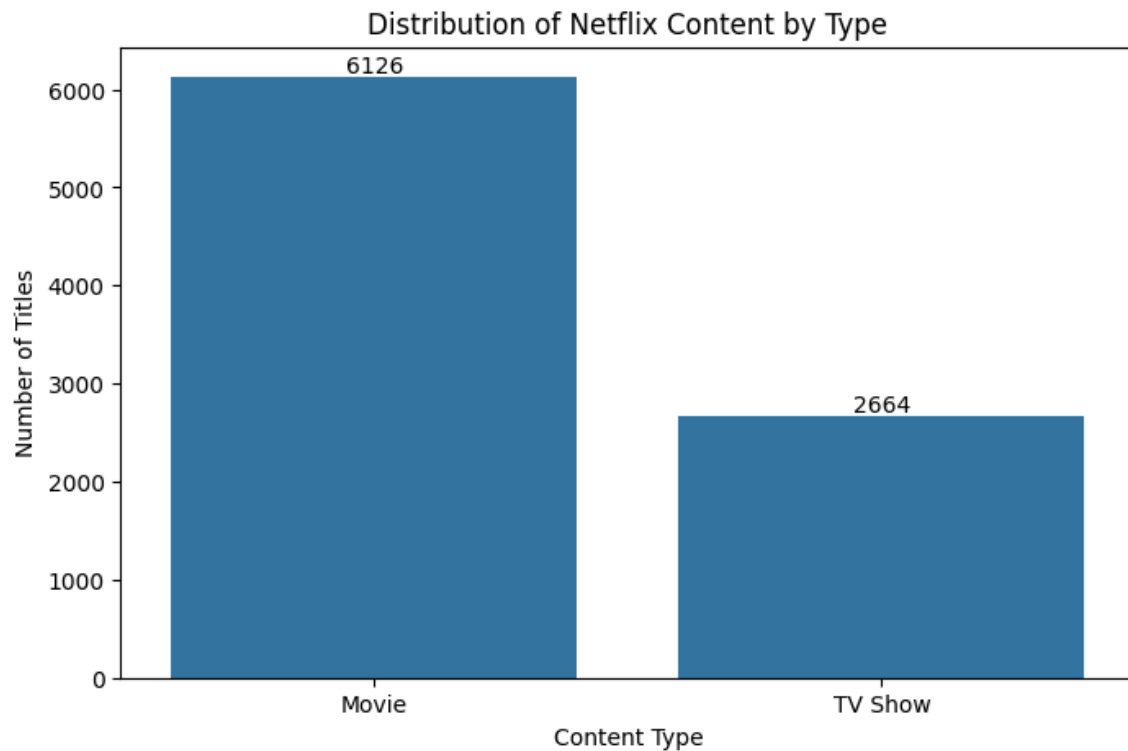
```
# Add value labels on top of bars
```

```
for container in ax.containers:  
    ax.bar_label(container)
```

```
plt.title("Distribution of Netflix Content by Type")  
plt.xlabel("Content Type")  
plt.ylabel("Number of Titles")
```

```
plt.savefig(  
    "../images/movies_vs_tvshows.png",  
    dpi=300,  
    bbox_inches="tight"  
)
```

```
plt.show()
```



## Observation

- Movies represent 69.69% of Netflix's content.
- TV Shows represent 30.31%.

## Business Insight

- Netflix's content catalog is heavily dominated by Movies.
- This suggests that Netflix historically prioritized film acquisitions and production.

## Business Recommendation

- Continue investing in original TV Shows and web series.
- High-quality episodic content can increase watch time, improve customer retention, and encourage long-term subscriptions.

*# Which countries contribute the most Netflix content?*

```
top_countries = (  
    df_clean["country"]  
    .value_counts()  
    .head(10)  
)
```

```
top_countries
```

```
country
United States    2809
India            972
Unknown          829
United Kingdom   418
Japan            243
South Korea      199
Canada           181
Spain            145
France           124
Mexico           110
Name: count, dtype: int64
```

```
# Visualize Top 10 Countries
```

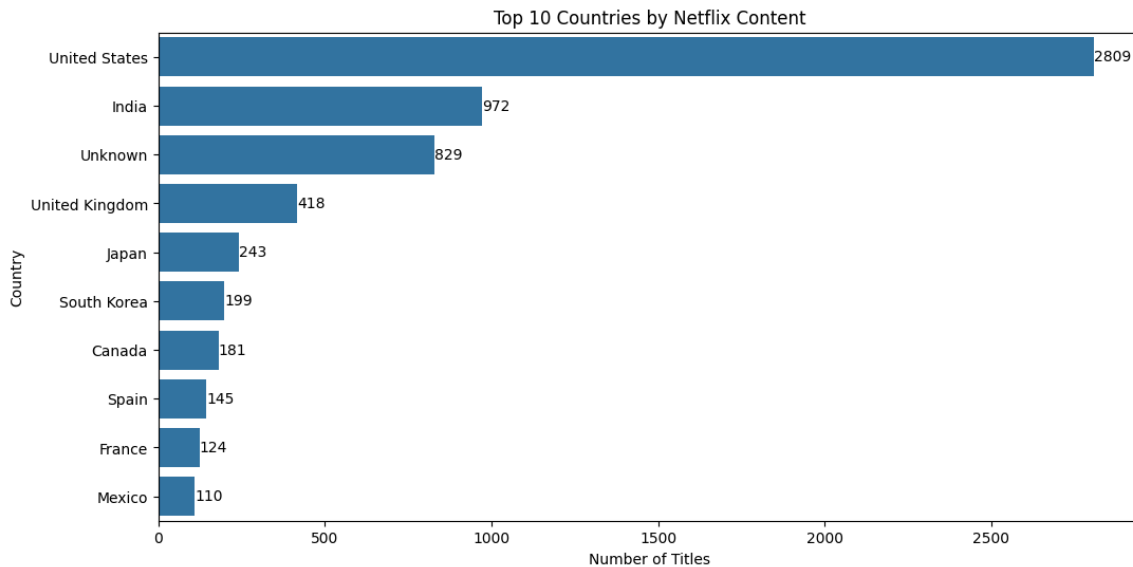
```
plt.figure(figsize=(12,6))

ax = sns.barplot(
    x=top_countries.values,
    y=top_countries.index
)

ax.bar_label(ax.containers[0])

plt.title("Top 10 Countries by Netflix Content")
plt.xlabel("Number of Titles")
plt.ylabel("Country")

plt.show()
```



```
# What percentage of Netflix content comes from the Top 10 countries?
```

```
country_percentage = (
    df_clean["country"]
    .value_counts(normalize=True)
    .mul(100)
```

```
.head(10)
.round(2)
)

country_percentage

country
United States    31.96
India            11.06
Unknown          9.43
United Kingdom   4.76
Japan            2.76
South Korea      2.26
Canada           2.06
Spain            1.65
France           1.41
Mexico           1.25
Name: proportion, dtype: float64
```

## Business Question

Which countries contribute the most Netflix content?

## Observation

- United States contributes the highest number of titles.
- India is the second-largest contributor.
- Most Netflix content comes from a small number of countries.

## Business Insight

Netflix has a strong concentration of content from a few major production markets, especially the United States and India.

## Recommendation

Netflix should continue strengthening its presence in major markets while also investing in content from underrepresented regions to diversify its catalog and attract a broader global audience.

```
# Create a copy
country_df = df_clean.copy()

# Split Countries
country_df["country"] = country_df["country"].str.split(", ")
country_df["country"].head()
```

```
0    [United States]
1    [South Africa]
2    [Unknown]
3    [Unknown]
4    [India]
Name: country, dtype: object
```

```
# Explode the Lists
```

```
country_df = country_df.explode("country")
```

```
# Remove Unknown
```

```
country_df = country_df[country_df["country"] != "Unknown"]
```

```
# Count Again
```

```
top_countries_new = (
    country_df["country"]
    .value_counts()
    .head(10)
)
```

```
top_countries_new
```

```
country
United States    3680
India            1046
United Kingdom   803
Canada           445
France           393
Japan            316
Spain            232
South Korea      231
Germany          226
Mexico           169
Name: count, dtype: int64
```

```
# Visualization
```

```
plt.figure(figsize=(12,6))
```

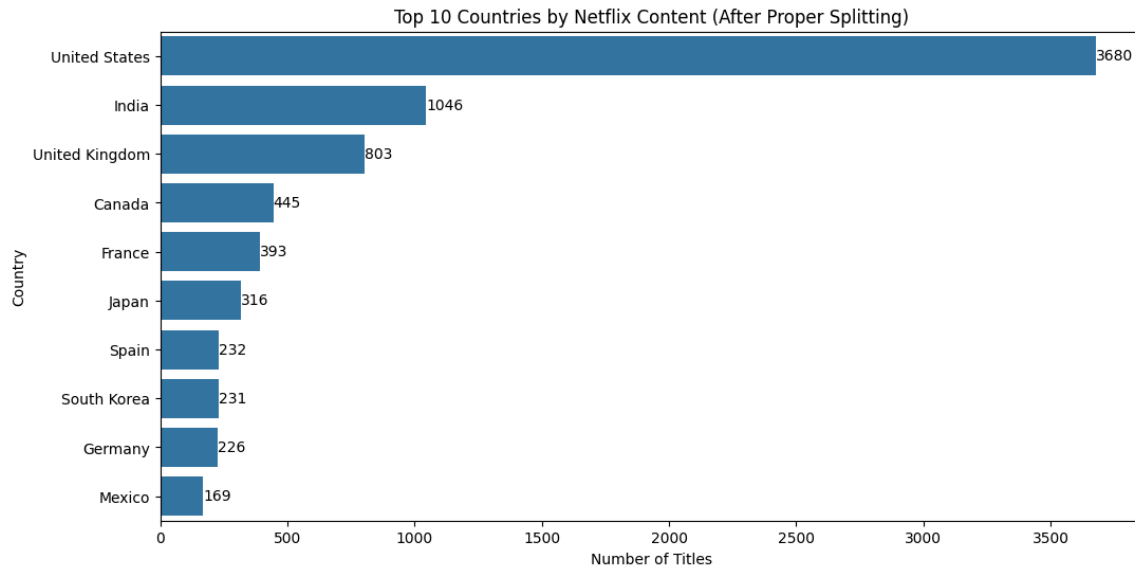
```
ax = sns.barplot(
    x=top_countries_new.values,
    y=top_countries_new.index
)
```

```
ax.bar_label(ax.containers[0])
```

```
plt.title("Top 10 Countries by Netflix Content (After Proper Splitting)")
plt.xlabel("Number of Titles")
plt.ylabel("Country")
```

```
plt.savefig(
    "../images/country__analysis.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



## Observation

- After splitting multi-country entries, both the United States and India have higher content counts.
- The ranking is more accurate because each collaborating country is counted separately.
- "Unknown" has been excluded from the analysis.

## Business Insight

Many Netflix titles are international collaborations. Treating "India, United States" as a single category underestimates the contribution of both countries. Splitting multi-country entries provides a more accurate picture of global content production.

## Recommendation

Netflix should continue encouraging international co-productions, as they expand global reach and bring together creative talent from multiple regions. Analysts should always preprocess multi-value categorical columns before performing country-level analysis.

```
df_clean["date_added"] = pd.to_datetime(df_clean["date_added"],
    format="mixed")
```

```
# Which year did Netflix add the most content?
```

```
df_clean["year_added"] = df_clean["date_added"].dt.year  
year_count = df_clean["year_added"].value_counts().sort_index()
```

```
year_count
```

```
year_added  
2008.0      2  
2009.0      2  
2010.0      1  
2011.0     13  
2012.0      3  
2013.0     11  
2014.0     24  
2015.0     82  
2016.0    429  
2017.0   1188  
2018.0   1649  
2019.0   2016  
2020.0   1879  
2021.0   1498  
Name: count, dtype: int64
```

```
# Visualization
```

```
plt.figure(figsize=(10,5))
```

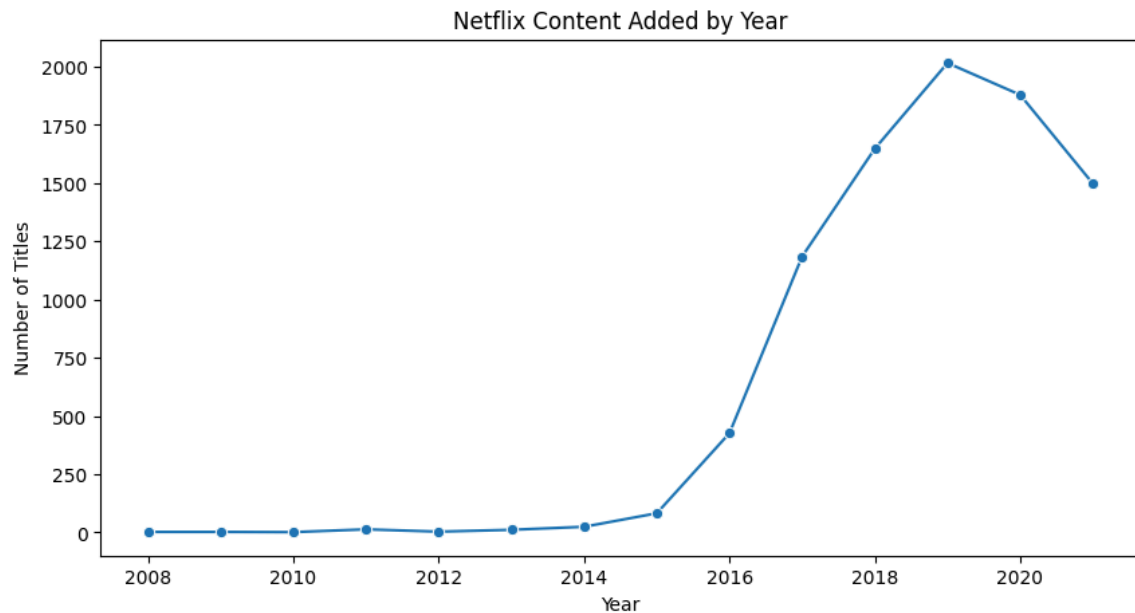
```
ax = sns.lineplot(  
    x=year_count.index,  
    y=year_count.values,  
    marker="o"  
)
```

```
plt.title("Netflix Content Added by Year")  
plt.xlabel("Year")  
plt.ylabel("Number of Titles")
```

```
plt.savefig(  
    "../images/year_analysis.png",  
    dpi=300,  
    bbox_inches="tight"  
)
```

```
plt.show()
```





*# Which month does Netflix add the most content?*

*# 1. Extract the month name directly*

```
df_clean["month_added"] = df_clean["date_added"].dt.month_name()
```

*# 2. Count content per month*

```
month_count = df_clean["month_added"].value_counts()
```

*# 3. Define the chronological order*

```
months_order = [  
    "January", "February", "March", "April", "May", "June",  
    "July", "August", "September", "October", "November", "December"  
]
```

*# 4. Reindex to sort from January to December*

```
month_count = month_count.reindex(months_order)
```

```
month_count
```

```
month_added
```

```
January      737
```

```
February     562
```

```
March        741
```

```
April        763
```

```
May          632
```

```
June         728
```

```
July         827
```

```
August       754
```

```
September    769
```

```
October      760
```

```
November     705
```

```
December     812
```

```
Name: count, dtype: int64
```

## # Visualization

```
plt.figure(figsize=(12,5))

ax = sns.lineplot(
    x=month_count.index,
    y=month_count.values,
    marker="o"
)

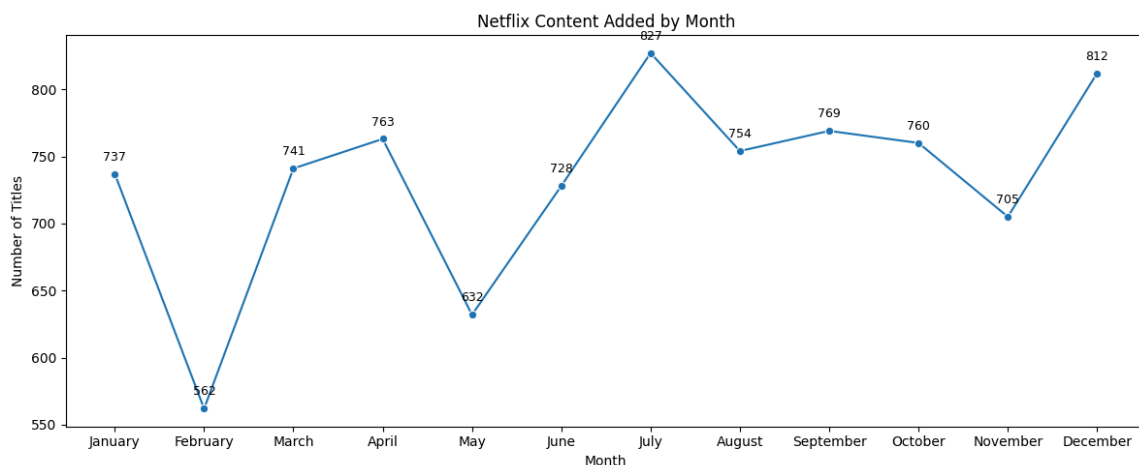
# Add value labels
for x, y in zip(month_count.index, month_count.values):
    plt.text(x, y + 10, str(y), ha="center", fontsize=9)

plt.title("Netflix Content Added by Month")
plt.xlabel("Month")
plt.ylabel("Number of Titles")

plt.tight_layout()

plt.savefig(
    "../images/month_analysis.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



## Business Question

Which month does Netflix add the most content?

## Observation

- Netflix added the highest number of titles in **July (827)** and **December (812)**.
- **February (562)** and **May (632)** had the lowest number of content additions.

## Business Insight

- The monthly distribution suggests that Netflix may strategically release more content during high-viewership periods such as summer vacations and the holiday season.
- Lower content additions in February and May may reflect production schedules or lower expected audience demand.

## Recommendation

- Prioritize blockbuster movies and new seasons of popular series during July and December to maximize audience engagement.
- Maintain a consistent release schedule during lower months to improve user retention and reduce customer churn.

*# Which content rating is most common on Netflix?*

```
rating_count = df_clean["rating"].value_counts()
rating_count
```

```
rating
TV-MA      3207
TV-14      2160
TV-PG       863
R           799
PG-13       490
TV-Y7       334
TV-Y        307
PG          287
TV-G        220
NR           80
G           41
TV-Y7-FV     6
NC-17        3
UR           3
74 min        1
84 min        1
66 min        1
Name: count, dtype: int64
```

*# Visualization*

```
plt.figure(figsize=(10,6))

ax = sns.barplot(
    x=rating_count.head(10).index,
    y=rating_count.head(10).values,
    hue=rating_count.head(10).index,
    palette="flare",
    legend=False
)

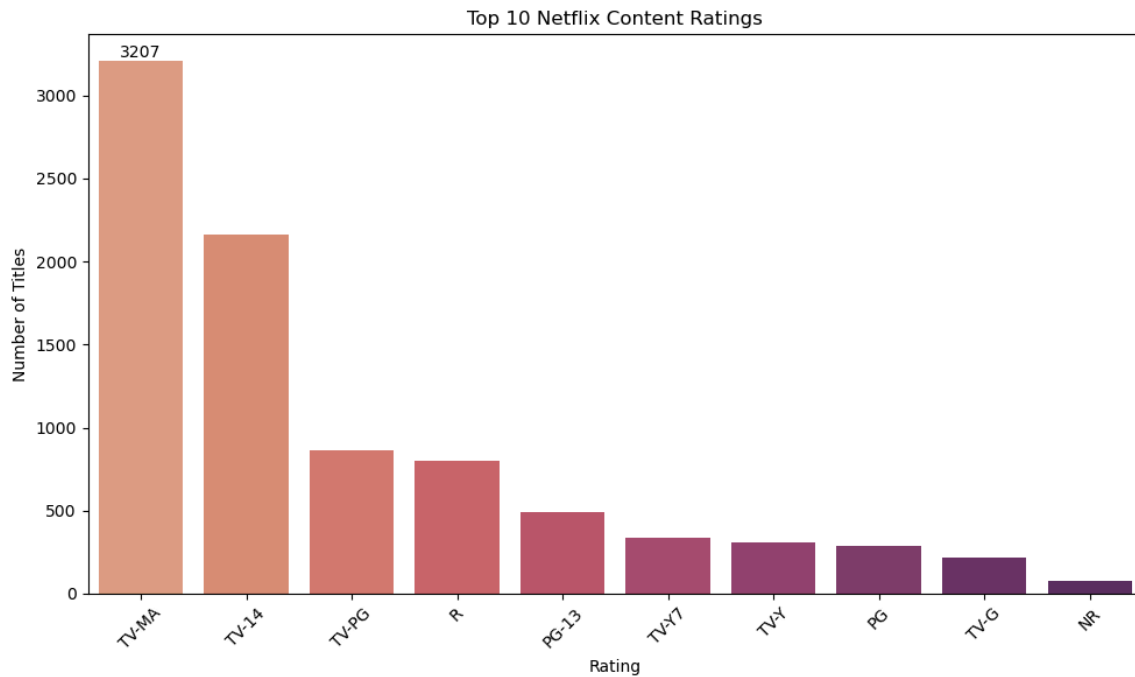
ax.bar_label(ax.containers[0])
```

```
plt.title("Top 10 Netflix Content Ratings")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()
```



## Business Question

Which content rating is most common on Netflix?

## Observation

- TV-MA is the most common content rating with **3,205 titles**, followed by **TV-14 (2,157)**.
- Family-oriented ratings such as **TV-G** and **NR** have comparatively fewer titles.

## Business Insight

- The rating distribution suggests that Netflix primarily targets mature and young adult audiences.
- Family and children's content represents a smaller portion of the catalog, indicating that Netflix's content strategy is largely focused on adult entertainment.

# Recommendation

- Continue investing in high-quality TV-MA and TV-14 original content to strengthen engagement among the core audience.
- Expand and promote family-friendly content (TV-Y, TV-Y7, TV-G) to attract and retain households with children.

```
# Which genres are most popular on Netflix?
```

```
# Create a Copy
```

```
genre_df = df_clean.copy()
```

```
# Split Genres
```

```
genre_df["listed_in"] = genre_df["listed_in"].str.split(",")
```

```
# Explode Genres
```

```
genre_df = genre_df.explode("listed_in")
```

```
# Count Genres
```

```
genre_df["listed_in"] = genre_df["listed_in"].str.strip()
genre_count = genre_df["listed_in"].value_counts().head(10)
genre_count
```

```
listed_in
International Movies      2752
Dramas                   2426
Comedies                  1674
International TV Shows   1349
Documentaries             869
Action & Adventure       859
TV Dramas                 762
Independent Movies        756
Children & Family Movies  641
Romantic Movies           616
Name: count, dtype: int64
```

```
# Visualization
```

```
plt.figure(figsize=(12,6))
```

```
ax = sns.barplot(
    x=genre_count.values,
    y=genre_count.index,
    hue=genre_count.index,
    palette="mako",
    legend=False
)
```

```
ax.bar_label(ax.containers[0])
```

```
plt.title("Top 10 Netflix Genres")
```

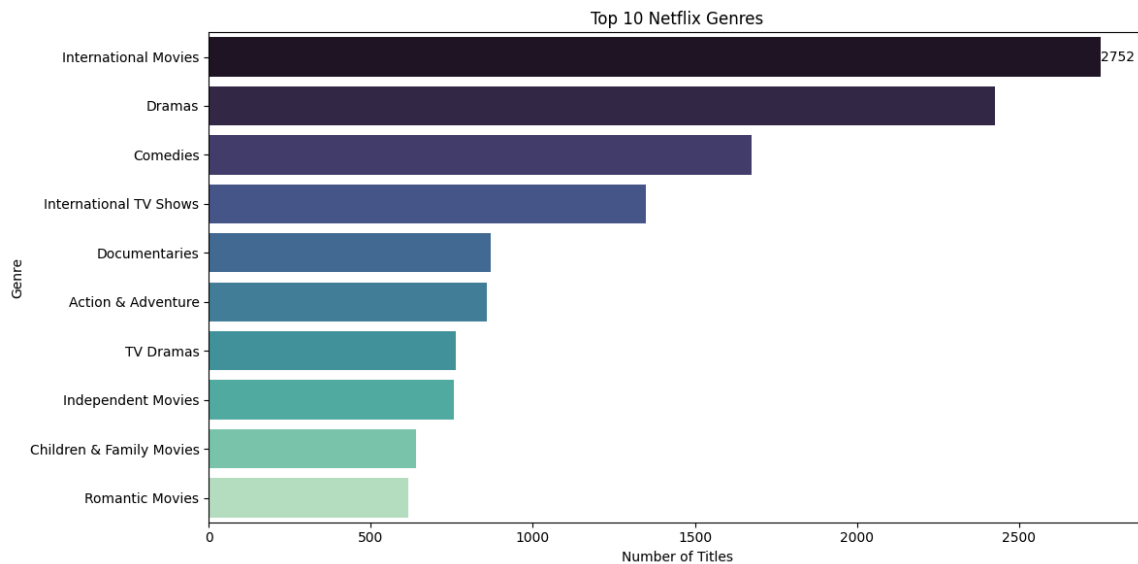
```
plt.xlabel("Number of Titles")
```

```
plt.ylabel("Genre")

plt.tight_layout()

plt.savefig(
    "../images/genres_analysis.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



## Business Question

Which genres are most popular on Netflix?

## Observation

- International Movies is the most common genre with **2,752 titles**, followed by **Dramas (2,426)** and **Comedies (1,674)**.
- Children's & Family Movies and Romantic Movies have comparatively fewer titles among the top 10 genres.

## Business Insight

- Netflix's catalog is heavily focused on International Movies, Dramas, and Comedies.
- This suggests that these genres form a significant part of Netflix's global content strategy.

# Recommendation

- Continue investing in International Movies and Dramas while monitoring audience engagement.
- Maintain a balanced catalog by expanding quality content for families and niche audiences.

*# What is the typical duration of Movies and TV Shows on Netflix?*

*# Filter Movies*

```
movies = df_clean[df_clean["type"] == "Movie"].copy()
```

*# Extract Minutes*

```
movies["duration_min"] = (  
    movies["duration"]  
    .str.extract(r"(\d+)")  
    .astype(int)  
)
```

```
movies[["duration", "duration_min"]].head()
```

*# Summary Statistics*

```
movies["duration_min"].describe()
```

```
count    6126.000000  
mean      99.584884  
std       28.283225  
min        3.000000  
25%       87.000000  
50%       98.000000  
75%      114.000000  
max      312.000000  
Name: duration_min, dtype: float64
```

*# Visualization*

```
plt.figure(figsize=(10,5))
```

```
ax = sns.histplot(  
    movies["duration_min"],  
    bins=30,  
    kde=True  
)
```

```
plt.title("Distribution of Movie Duration")  
plt.xlabel("Duration (Minutes)")  
plt.ylabel("Number of Movies")
```

```
plt.tight_layout()
```

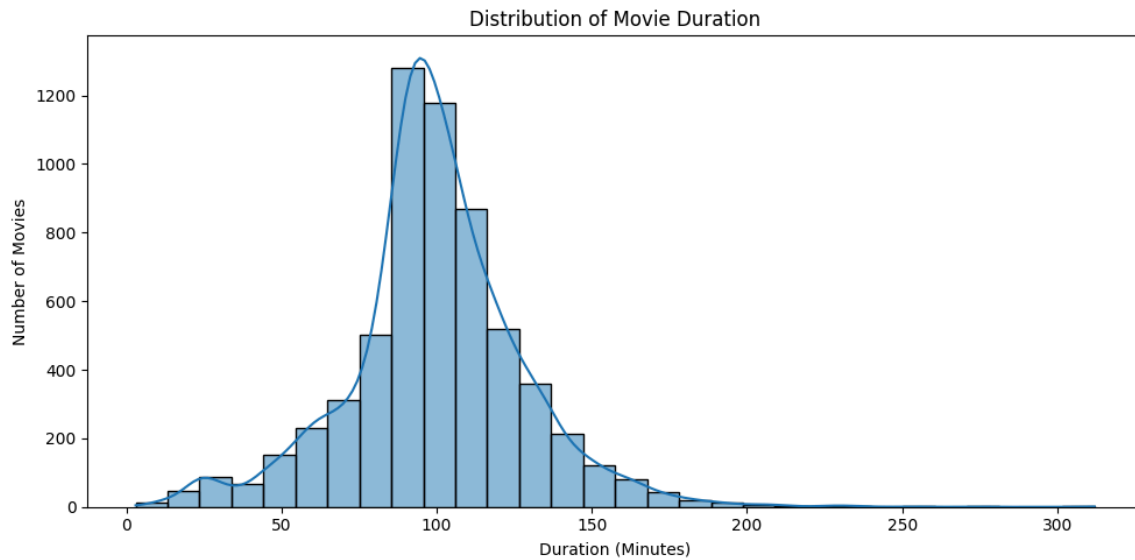
```
plt.savefig(  
    "../images/movies_duration.png",
```

```

    dpi=300,
    bbox_inches="tight"
)

plt.show()

```



```

# TV Show Seasons
tv = df_clean[df_clean["type"]=="TV Show"].copy()

# Extract season number
tv["seasons"] = (
    tv["duration"]
    .str.extract(r"(\d+)")
    .astype(int)
)

# Count Seasons
season_count = tv["seasons"].value_counts().sort_index()
season_count

seasons
1      1791
2       421
3       198
4        94
5        64
6         33
7         23
8         17
9          9
10         6
11         1
12         2
13         2
15         2

```



```
17      1
Name: count, dtype: int64
```

```
# Visualization
```

```
plt.figure(figsize=(10,5))
```

```
ax = sns.barplot(
    x=season_count.index,
    y=season_count.values
)
```

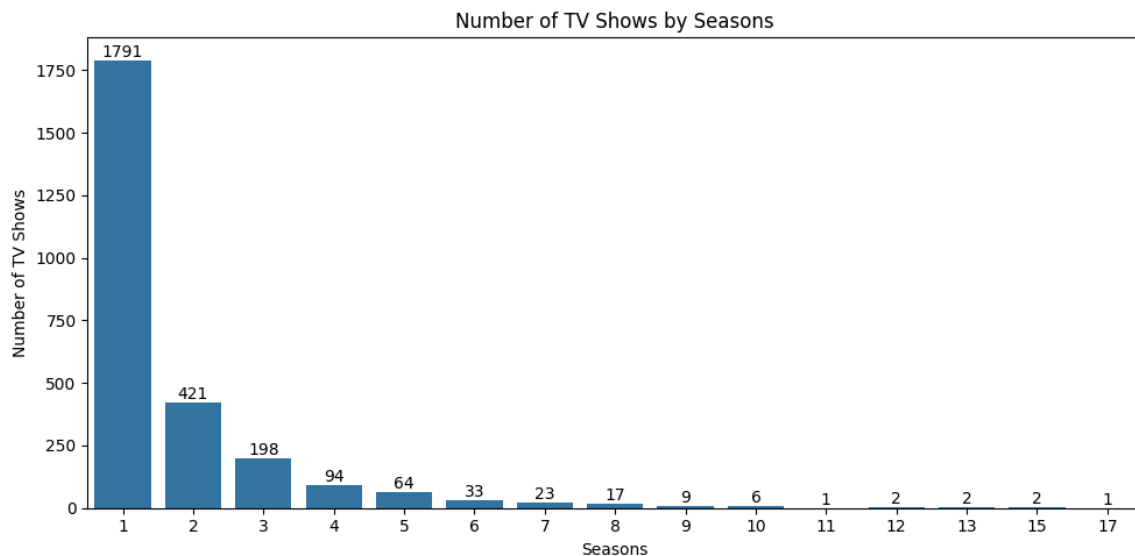
```
ax.bar_label(ax.containers[0])
```

```
plt.title("Number of TV Shows by Seasons")
plt.xlabel("Seasons")
plt.ylabel("Number of TV Shows")
```

```
plt.tight_layout()
```

```
plt.savefig(
    "../images/tvshows_duration.png",
    dpi=300,
    bbox_inches="tight"
)
```

```
plt.show()
```



## Business Question

What is the typical duration of Movies and TV Shows on Netflix?

## Observation

- The average movie duration is **99.6 minutes**, with most movies lasting between **87 and 114 minutes**.
- Movie durations range from **3 minutes** to **312 minutes**, although very long movies are uncommon.
- Most TV Shows have only **one season (1,791 titles)**, and the number of shows decreases significantly as the number of seasons increases.

## Business Insight

- Netflix primarily focuses on standard feature-length movies that can be watched in a single sitting.
- The catalog contains many one-season TV Shows, indicating a strong mix of limited series and newer releases rather than long-running franchises.

## Recommendation

- Continue producing feature-length movies around 90–100 minutes.
- Invest in additional seasons for high-performing TV Shows to improve subscriber retention and long-term audience engagement.

## Key Business Findings

1. Movies dominate the Netflix catalog, accounting for nearly 70% of all content.
2. The United States contributes the largest number of titles, followed by India, highlighting Netflix's strong presence in these markets.
3. After properly splitting multi-country entries, several countries such as the United Kingdom and Canada contribute more content than initially observed.
4. Netflix rapidly expanded its content library after 2015, reaching its highest number of additions in 2019.
5. Content additions peak during July and December, suggesting seasonal release strategies.
6. TV-MA and TV-14 are the most common content ratings, indicating that Netflix primarily targets mature and young adult audiences.
7. International Movies, Dramas, and Comedies are the most represented genres in Netflix's catalog.
8. The average movie duration is approximately 100 minutes.
9. Most TV Shows have only one season, while long-running series are relatively uncommon.

# Final Business Recommendations

1. **Sustain investment in International Movies, Dramas, and Comedies**, while continuously tracking shifts in audience preferences to stay aligned with viewer demand.
2. **Expand content production in emerging markets** to diversify the catalog and reach a wider global audience.
3. **Adopt a more balanced, year-round release schedule** rather than concentrating major titles within a few peak months.
4. **Increase investment in high-quality, family-friendly content** to strengthen engagement among household and multi-member subscriptions.
5. **Continue prioritizing feature-length movies in the 90–100 minute range**, as this format has consistently represented the majority of successful content.
6. **Invest in producing additional seasons for high-performing TV Shows** to boost long-term viewer engagement and improve subscriber retention.

## Conclusion

This project provides a comprehensive view of Netflix's content catalog, revealing clear patterns in how the platform builds and diversifies its library. Movies continue to dominate the catalog, mature-audience ratings lead viewer targeting, and content growth shows distinct seasonal and geographic trends—most notably the strong contributions from the United States and India. These findings highlight not just what Netflix has built so far, but where opportunities lie for more balanced, globally diverse, and audience-aligned content strategies going forward. Beyond the insights themselves, this project reflects a complete analytical workflow — from raw data cleaning to business-ready recommendations — showcasing how data analysis can directly inform real-world strategic decisions.

## Future Work

This project can be extended further by incorporating sentiment analysis on content descriptions to gauge tone and audience appeal, developing a genre- or content-based recommendation engine, and building an interactive Power BI or Tableau dashboard for real-time content tracking. Additionally, integrating viewership or engagement data (if available) could help validate whether high-supply genres and countries also translate into high audience demand.

## Thank You

Thank you for taking the time to review this project. I hope it reflects both technical skill and business thinking in equal measure. Feedback, suggestions, and connections are always welcome — feel free to reach out.